

DATA STRUCTURE LAB

ADT ARRAY

Integer Array

INTEGER ARRAY

"Write a C program to demonstrate pointer arithmetic on an integer array. Display element values using both array indexing and pointer notation. Show address difference between consecutive elements. Sample Input: Enter number of elements: 4 Enter elements: 10 20 30 40"

```
2 int main(){
3     int n,i;
4     //printf("enter number of elements: ");
5     scanf("%d",&n);
6     int arr[n];
7     int *ptr=arr;
8     for(i=0;i<n;i++){
9         scanf("%d",&arr[i]);
10    }
11    printf("\ndisplay using array indexing:\n");
12    for(i=0;i<n;i++){
13        printf("arr[%d]=%d\n",i,arr[i]);
14    }
15    printf("\ndisplay using pointer notation:\n");
16    for(i=0;i<n;i++){
17        printf("*(ptr+%d)=%d\n",i,*(ptr+i));
18    }
19    printf("the address difference between arr[0] and arr[1] is %ld", (char*)&arr[1]-(char*)&arr[0]);
20    return 0;
21 }
```

OUTPUT

display using array indexing:
arr[0]=10
arr[1]=20

4
10
20
30
40

Windows

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ENG IN

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion. Sample Input: Enter a number- 6 Sample Output: Factorial of 6 is 720"

```
1 #include<stdio.h>
2 int fact(int n)
3 {
4     if(n==1)
5         return 1;
6
7     return n*fact(n-1);
8 }
9 int main()
10 {
11     int n;
12     scanf("%d",&n);
13     printf("%d",fact(n));
14     return 0;
15 }
```

OUTPUT

120

5

Windows

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DS PROGRAMMING

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QUESTIONS 8 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K ($0 \leq K < N$), show original→rotated→reversal steps visually, compute the sum difference, and output results."

PROGRAM EDITOR

C
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```

1 #include<stdio.h>
2 void reverse(int a[], int start, int end)
3 {
4     int temp;
5     while(start<end)
6     {
7         temp=a[start];
8         a[start]=a[end];
9         a[end]=temp;
10        start++;
11        end--;
12    }
13 }
14 void display(int a[],int n)
15 {
16     int i;
17     printf("[");
18     for(i=0;i<n;i++)
19         printf("%d",a[i]);
20     printf("]\n");
21 }

```

OUTPUT

```

get the numbers: declare the positionoriginal: [12345678910]

```

INPUT

```

1 2 3 4 5 6 7 8 9 10
3

```

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QUESTIONS 8 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; input sparse...

PROGRAM EDITOR

C
Run
Save

```

1 #include <stdio.h>
2
3 typedef struct {
4     int row;
5     int col;
6     int val;
7 } Triple;
8
9 int main() {
10     int r1, c1, nz1, r2, c2, nz2;
11     Triple m1[100], m2[100], sum[200];
12     int sum_count = 0;
13
14     // Input Matrix 1
15     if (scanf("%d %d %d", &r1, &c1, &nz1) != 3) return 0;
16     for (int i = 0; i < nz1; i++) {
17         scanf("%d %d %d", &m1[i].row, &m1[i].col, &m1[i].val);
18     }
19
20     // Input Matrix 2
21

```

OUTPUT

```

0 8 0

```

INPUT

```

3 3 3
0 1 5
1 0 8
2 2 2
3 3 2
0 1 3
2 2 7

```



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QUESTIONS 8 / 10 completed

Fibonacci Series

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series (n ≤ 20) without using recursion. Start with 0 and 1.
Sample Input/Output:
Enter number of terms: 8
Fibonacci series: 0 1 1 2 3 5 8 13

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a=0,b=1,c,i;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++){
7         printf("%d",a);
8         c=a+b;
9         a=b;
10        b=c;
11    }
12    return 0;
13 }
```

OUTPUT

0 1 1 2 3

INPUT

5



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QUESTIONS 8 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int i,n;
4     scanf("%d",&n);
5     int a[n]; //declare array with size n
6     for(i=0;i<n;i++){
7         scanf("%d",&a[i]);
8     }
9
10    printf("%d",a[4]);
11    return 0;
12 }
```

OUTPUT

INPUT

6
10 20 30 40 50 60

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Fibonacci series (N ▾)

FIBONACCI SERIES (N ≤ 50)

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Sample Input/Output:
text
Enter number of terms (1-50): 10
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
Sum of series: 88
Largest term: 34
Enter number of terms (1-50): 60
Invalid input! Please enter terms between 1 and 50.

PROGRAM EDITOR

C ▾ ✓ Updated Run Save

```
29         next = a + b;
30     }
31
32     fib[i] = next;
33     sum += next;
34     if (next > largest) largest = next;
35
36     a = b;
37     b = next;
38 }
39
40 printf("Fibonacci Series: ");
41 for (int i = 0; i < N; i++) {
42     printf("%lld", fib[i]);
43     if (i < N - 1) printf(" ");
44 }
45 printf("\n");
46
47 printf("Sum of series: %lld\n", sum);
48 printf("Largest term: %lld\n", largest);
49
```

OUTPUT

Enter number of terms (1-50): Fibonacci Series: 0 1 1 2 3 5 8 13 21 34

INPUT

10

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QUESTIONS 8 / 10 completed

Unsorted Array ▾

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.
Sample Input: a=[10,9,7,4,3,1,2]
Sample output: 5,6,8

PROGRAM EDITOR

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```
1 #include<stdio.h>
2 int main(){
3     int a[]={10,9,7,4,3,1,2};
4     int n=7;
5     int i,j,found;
6     int max=a[0];
7     for(i=1;i<n;i++){
8         if(a[i]>max)
9             max=a[i];
10    }
11    printf("missing elements: ");
12    for(i=1;i<=max;i++){
13        found = 0;
14        for(j=0;j<n;j++){
15            if(a[j]==i){
16                found = 1;
17                break;
18            }
19        }
20        if(found==0){
21            printf("%d ",i);
22        }
23    }
24 }
```

OUTPUT

missing elements: 5 6 8

INPUT

Your INPUT goes here!



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QUESTIONS 8 / 10 completed

Fib series-iterative

FIB SERIES-ITERATIVE & RECUR

calls made.

Sample Input:
Enter N: 7

Sample Output:
Iterative: 0 1 1 2 3 5 8
Recursive: 0 1 1 2 3 5 8
Recursive Calls: 13
Iterative Time: 0.001 ms
Recursive Time: 0.005 ms

Test Case

PROGRAM EDITOR

C

Run

Save

```
42
43 // Recursive
44 recursiveCalls = 0;
45 clock_t startRec = clock();
46 printf("\nRecursive: ");
47 for (int i = 0; i < N; i++) {
48     printf("%lld", fibRec(i));
49     if (i != N - 1) printf(" ");
50 }
51 clock_t endRec = clock();
52
53 printf("\nRecursive Calls: %lld", recursiveCalls);
54
55 double iterTime = timeTaken(startIter, endIter);
56 double recTime = timeTaken(startRec, endRec);
57
58 printf("\nIterative Time: %.3f ms", iterTime);
59 printf("\nRecursive Time: %.3f ms", recTime);
60
61 return 0;
62 }
```

OUTPUT

Enter N: Iterative: 0 1 1 2 3 5 8

INPUT

7



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QUESTIONS 10 / 10 completed

Structure

STRUCTURE

Sample Input:
Enter number of students: 2
101 Ravi 75
102 Kiran 85

Sample Output:
Highest Mark: 85 (Kiran)
Class Average: 80.00
2 records stored successfully.

Test Case:

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define NAME_LEN 50
5
6 typedef struct {
7     int rollNo;
8     char name[NAME_LEN];
9     float marks;
10 } Student;
11
12 int main() {
13     int n, i;
14     Student arr[100];
15
16     printf("Enter number of students: ");
17     scanf("%d", &n);
18
19     for (i = 0; i < n; i++) {
20         printf("Enter Roll No, Name, Marks: ");
21         scanf("%d %s %f", &arr[i].rollNo, arr[i].name, &arr[i].marks);
```

OUTPUT

Enter number of students: Enter Roll No, Name, Marks: Enter Roll No, Name, Marks: Highest Mark: 102 (ravi75)

INPUT

2
101 ravi75
102 kiran 85